The problem of Alzheimer's disease as a clue to immortality Part 2

V. HORMONE IMBALANCE, LEADING TO FAILURE OF PROTECTIVE INHIBITION AND ALZHEIMER'S DISEASE

"All cell death is characterized by an increase of intracellular calcium...." "Increase of cytoplasmic free calcium may therefore be called 'the final common path' of cell disease and cell death. Aging as a background of diseases is also characterized by an increase of intracellular calcium. Diseases typically associated with aging include hypertension, arteriosclerosis, diabetes mellitus and dementia."

T. Fujita, "Calcium, parathyroids and aging," in Calcium-Regulating Hormones. 1. Role in Disease and Aging, H. Morii, editor, Contrib. Nephrol. Basel, Karger, 1991, vol. 90, pp. 206-211.

THE FUNCTION OF ENERGY

Most people are slightly demented now and then, when they are very sleepy or tired, or sick, or drunk, or having a hormone imbalance or extreme anxiety state. Sometimes physicians have described people as demented, implying that the condition would never improve, when the person was depressed or hypothyroid. If the person has a history of epilepsy, or is very old, the physician is more likely to diagnose dementia than if the same loss of mental function occurs in a younger person without a history of a nervous disorder. Even people with less education are at increased risk of being diagnosed as "demented."

In 1976, I saw a 52 year-old woman who had the diagnosis of epileptic dementia. After 3 or 4 days of taking progesterone, her mental function returned to the extent that she could find her way around town by herself, and could work. A few months later, she returned to graduate school, got straight As and a master's degree. A few years later, a man in his 80s showed the classical signs of senile dementia, with childishness, confusion, self-centeredness, and unstable emotions. A few days after getting a mixture of thyroid, pregnenolone, and progesterone, his mind was again clear, and he was able to work on a research project he had set aside years before.

When the body temperature is very much below normal, mental functioning is seriously limited. I think the first question that should be asked about a demented person is "is this the cold brain syndrome, or is something else involved?" When it is known that the brain has shrunken drastically, and filled up with plaques and developed gliosis, we know that something more than a "cold brain" is involved, but we don't know how much function could be regained if the hormones were normalized. Every moment of malfunction probably leaves its structural mark. Early or late, it is good to prevent the functional errors that lead to further damage, and to give the regenerative systems an opportunity to work. Before the final "calcium death" described (above) by Fujita, there are many opportunities for intervening to stop or reverse the process. The older the person is, the more emphasis should be put on protective inhibition, rather than immediately increasing energy production. Magnesium, carbon dioxide, sleep, red light, and naloxone might be appropriate at the beginning of therapy.

The resting state of a cell is a highly energized state. To the old Pavlovians, the resting state existed at two energy levels, and they applied the term "protective inhibition" generally to the depleted state (parabiosis) that occurs in exhaustion or coma, but I am using the phrase in a more general sense, that seems reasonable now that the concept of "excitotoxic" injury has become current. I mean it to include everything which protects against excitotoxic injury. This definition therefore has the virtue of being biochemically and physiologically very specific, while retaining the functional and therapeutic significance that it had for the Pavlovians. (My book, Mind and Tissue, and the chapter "A unifying principle" in Generative Energy, discussed the idea of the resting state and protective inhibition.)

Ordinary healthy sleep is an example of restorative, protective inhibition. The energy charge, including levels of ATP, creatine phosphate, and glycogen storage, regulates many restorative enzyme systems. I have suggested (1975, J. Orthomol. Psychiatry) how the entropy-sensitivity or cold-inactivation of an enzyme could be involved in shifting the brain toward a state of inhibition. A recent publication (J. H. Benington and H. C. Heller, "Restoration of brain energy metabolism as the function of sleep," Progress in Neurobiol. 45, 347-360, 1995) has proposed that reduction of energy charge and depolarization of cells act through adenosine secretion to restore glycogen stores. Since glycogen stores decrease with aging, this work supports the idea that protective inhibition is weakened with aging. (L. N. Simanovskiy and Zh. A. Chotoyev, "The effect of hypoxia on glycogenolysis and glycolysis rates in the rat brain," Zhurnal Evolyutsionnoy Biokhimii i Fiziologii 6(5), 577-579, 1970.)

J. H. Benington and H. C. Heller, "Restoration of brain energy metabolism as the function of sleep," Prog. in Neurobiology 45, 347-360, 1995. "...the conditions that have been demonstrated to stimulate adenosine release from neural tissue represent either increases in metabolic demand (...activation of excitatory receptors) or decreases in metabolic supply (hypoxia, ischaemia, hypoglycemia)...." "In the brain, adenosine-stimulated increases in potassium conductance produce hyperpolarization, thereby reducing neuronal responsiveness...." "Adenosine release is triggered globally in response to changes in cerebral energy homeostasis." "A number of findings provide indirect support for the hypothesis that glycogen stores are depleted during waking and restored during sleep." "Reduced availability of glycogen to astrocytes must...increase adenosine release...." "Because ATP concentration is 100-fold greater than AMP concentration, a minute decrease in cellular energy charge...is translated into a large proportional increase in extracellular adenosine cencentration...."

The terms "functional quiescence" and "G0 quiescence" are similar in meaning to the resting state; I think of cells in the state of "G0 quiescence" as being stem cells, waiting for use in regeneration, but I don't subscribe to the idea that they can't be reconstituted from functioning, differentiated cells. In plants, dedifferentiation is achieved fairly easily, and in the study of animal cells the trend in that direction seems very obvious, though many people keep saying that it just isn't possible. In general, the things such as lipid peroxidation or calcium influx which cause cell replication at one level, cause cell death at a higher level.

Energy to resist stress makes quiescence possible, and prevents the deterioration of cells, of the sort that occurs in aging. O. Toussaint, et al., "Cellular aging and energetic factors," Exp. Gerontology 30(1), 1-22, 1995. "Experiments performed with endothelial cells in the context of the ischemia-reperfusion toxicity of free radicals, also offer good examples of the impact of cell energy on cell resistance to these toxic molecules." "...if a supplement of energy is given...the toxic effect of the free radicals is much reduced...."

The specific approaches of this orientation --to energize but quiet the brain--are diametrically opposed to some of the "therapies" for Alzheimer's disease that have been promoted recently by the drug industries: Things to increase stimulation, especially to increase cholinergic excitation; even the excitotoxic amino acids themselves and their analogs; and estrogen, which is a multiple brain excitant, proconvulsant, excitotoxic promoter, and anti-memory agent. Those product-centered publications stand out distinctly from the actual research.

There are many energy-related vicious circles associated with aging, but the central one seems to be the fat-thyroid-estrogen-free -radical-calcium sequence, in which the ability to produce stabilizing substances including carbon dioxide and progesterone is progressively lost, increasing susceptibility to the unstable unsaturated fats.

EFFECTS OF ESTROGEN AND UNSATURATED FATTY ACIDS

Estrogen production is facilitated when tissue is cooler, and it lowers body temperature. Estrogen and the endorphins act together in many ways (including the behavior of estrus), and naloxone (the antagonist of morphine and the endorphins) raises body temperature and in other ways opposes estrogen. Naloxone has been found to improve the symptoms of demented people, and I have seen it quickly, and dramatically, improve the mental clarity of a 60 year old woman who had used estrogen. It, like clonidine (the anti-adrenaline drug), is a good candidate for controlling the hot flashes and other symptoms of menopause.

In various degenerative brain conditions, blood clotting has been implicated either as a cause or a complication. Many people are promoting unsaturated oils for their "anti-clotting" value, in spite of the older literature showing that they inhibit proteolytic enzymes and slow clot removal. Several newer publications have revealed other aspects of their involvement in thrombus formation. A. J. Honour, et al., "The effects of changes in diet on lipid levels and platelet thrombus formation in living blood vessels," Br. J. Expt. Pathol. 59(4), 390-394, 1978--corn oil caused platelets to be more sensitive to ADP.

Although there is a lot of talk about "membrane fluidity," as a desirable thing, and the loss of unsaturated lipids in the aged brain, there are some interesting observations related to "viscosity" in Alzheimer's disease. The platelets of Alzheimer's patients are less viscous, and lipids extracted from the brain are more fluid, and contain 30% less cholesterol than normal (on a molar basis, in relation to phospholipids). (G. S. Roth, et al., 1995.) In general, lipid peroxidation causes cellular viscosity to increase, apparently by causing cross-linking of proteins, but I think the significance of the decreased cholesterol relates to its significance as precursor to pregnenolone and progesterone, and to the known association with Alzheimer's disease of a variant form of the cholesterol transporter protein, ApoE, which I suppose is a slightly less stable molecular form that is more susceptible to malfunction in stress.

The extracellular matrix is a major factor in the function and stability of brain cells. (L. F. Agnati, et al., "The concept of trophic units in the central nervous system," Prog. in Neurobiol. 46, 561-574, 1995. Any factor producing edema tends to disrupt the extracellular matrix (Chan and Fishman, 1978, 1980, and L. Loeb, 1948.)

Seizures are known to be promoted by estrogen, by unsaturated fats, and by lipid peroxidation, and to cause an increase in the size of the free fatty acid pool in the brain. Prolonged seizures cause nerve damage in certain areas, especially the hippocampus, thalamus, and neocortex (Siesjo, et al., 1989). Dementia is known to be produced by prolonged seizures.

Prenatal exposure to estrogen, to oxygen deficiency, or to unsaturated fats decreases the size of the brain at birth. There is apparently a requirement for saturated fats during development (J. M. Bourre, N. Gozlan-Devillierre, O. Morand, and N. Baumann, "Importance of exogenous saturated fatty acids during brain development and myelination in mice," Ann. Biol. Anim., Biochim., Biophys. 19(1B), 172-180, 1979.

Under the influence of estrogen, or unsaturated fats, brain cells swell, and their shape and interactions are altered. Memory is impaired by an excess of estrogen. Estrogen and unsaturated fat and excess iron kill cells by lipid peroxidation, and this process is promoted by oxygen deficiency. The fetus and the very old have high levels of iron in the cells. Estrogen increases iron uptake. Estrogen treatment produces elevation of free fatty acids in the blood, and lipid peroxidation in tissues. This tends to accelerate the accumulation of lipofuscin, age-pigment. Lactic acid, the production of which is promoted by estrogen, lowers the availability of carbon dioxide, leading to impairment of blood supply to the brain.

Estrogen stimulates cell division, but can also increase the rate of cell death. Unsaturated fatty acids can also stimulate or kill.

Both estrogen and unsaturated fats promote the formation of age-pigment. Besides increasing the free fatty acid concentration, estrogen possibly depresses the level of cholesterol, both of which are changes seen in the senile brain.

Estrogen causes massive alterations of extracellular matrix, and seems to promote dissolution of microtubules (Nemetschek-Gannsler), as calcium does. Unsaturated fats increase calcium uptake by at least some brain cells (H. Katsuki and S. Okuda, 1995.) Unsaturated fats, like estrogen, increase the permeability of blood vessels. The unsaturated fat causes edema of the brain, inhibits choline uptake, blocking acetylcholine production.

Progesterone is a nerve growth factor, produced by glial cells (oligodendrocytes). It promotes the production of myelin, protects against seizures, and protects cells against free radicals. It protects before conception, during gestation, during growth and puberty, and during aging. It promotes regeneration. Its production is blocked by stress, lipid peroxidation, and an excess of estrogen and iron.

Aspirin protects against iron toxicity, clot formation, and reduces lipid peroxidation while blocking prostaglandin formation. Aspirin and other antiinflammatory drugs, taken for arthritis, have been clearly associated with a reduced incidence of Alzheimer's disease. Aspirin reduces the formation of prostaglandins from arachidonic acid.

Unsaturated fatty acids, but not saturated fatty acids, are signals which activate cell systems.

Many different stimuli can induce cell activity, cell death, or change to another cell type. (J. Niquet, et al., "Glial reaction after seizure induced hippocampal lesion: Immunohistochemical characterization of proliferating glial cells," J. Neurocytol. 23(10), 641-656, 1994: "...hippocampal astrocytes from kainate-treated rats experess A2B5 immunoreactivity, a marker of type-2 astrocytes." "This suggests that in the CNS, normal resident astrocytes acquire the phenotypic properties of type-2 astrocytes.")

A "deficiency" of polyunsaturated fatty acids leads to altered rates of cellular regeneration and differentiation, a larger brain at birth, improved function of the immune system, decreased inflammation, decreased mortality from endotoxin poisoining, lower susceptibility to lipid peroxidation, increased basal metabolic rate and respiration, increased thyroid function, later puberty and decreases other signs of estrogen dominance. When dietary PUFA are not available, the body produces a small amount of unsaturated fatty acid (Mead acids), but these do not activate cell systems in the same way that plant-derived PUFAs do, and they are the precursors for an entirely different group of prostaglandins.

VITAMIN A AND THE STEROIDS

In a variety of cell types, vitamin A functions as an estrogen antagonist, inhibiting cell division and promoting or maintaining the functioning state. It promotes protein synthesis, regulates lysosomes, and protects against lipid peroxidation. Just as stress and estrogen-toxicity resemble aging, so does a vitamin A deficiency. While its known functions are varied, I think the largest use of vitamin A is for the production of pregnenolone, progesterone, and the other youth-associated steroids. One of vitamin E's important functions is protecting vitamin A from destructive oxidation. Although little attention has been given to the effects of unsaturated fats on vitamin A, their destruction of vitamin E will necessarily lead to the destruction of vitamin A. The increased lipid peroxidation of old age represents a vicious circle, in which the loss of the antioxidants and vitamin A leads to their further destruction.

To produce pregnenolone, thyroid, vitamin A, and cholesterol have to be delivered to the mitochondria in the right proportion and sufficient quantity. Normally, stress is balanced by increased synthesis of pregnenolone, which improves the ability to cope with stress. Lipid peroxidation, resulting from the accumulation of unsaturated fatty acids, iron, and energy deficiency, damages the mitochondrias' ability to produce pregnenolone. When pregnenolone is inadequate, cortisol is over-produced. When progesterone is deficient, estrogen's effect is largely unopposed. When both thyroid and progesterone are deficient, even fat cells synthesize estrogen.

THE NATURE OF ALZHEIMER'S DISEASE

Although Alzheimer's disease until recently referred to a certain type of organic dementia occuring in people in their thirties, forties and fifties (presenile dementia), structural similarities seen in senile dementia have caused the term to lose its original meaning. Alzheimer's sclerosis of blood vessels, and even the death of nerve cells, are sometimes neglected in favor of the more stylish ideas, emphasizing certain proteins that cause the tangles and plaques. Until recently, the "tangles" were commonly interpreted as the debris left after the death of a cell, rather than as one of the processes causing the death of the cell.

Alzheimer-type dementia is different from other dementias, but it overlaps with them, and with age-related and stress-related changes in other organs.

Physical signs (seen at autopsy) of AD:

1) Death of neurons (increase of glial cells),

2) Amyloid plaques (extracellular), associated with a particular variant of apolipoprotein E, the epsilon 4 allele,

3) Fibrillary tangles (intracellular, or remaining after the rest of the cell has disappeared),

4) Amyloid in blood vessels.

Functional and biochemical observations:

1) The mitochondrial energy problem, cytochrome oxidase and its regulation; body temperature/pulse-rate cycle disturbance; lipid peroxidation; respiratory defect; altered amino acid uptake; memory impairment; dominance of the excitatory systems vs. the inhibitory adenosine/GABA/progesterone /pregnenolone system. Increased calcium uptake, which is associated with lipid peroxidation and cell death. Increased cortisol and DHEA.

2) Deposit of abnormal proteins, such as transthyretin-amyloid; albumin binding of PUFA, vs. transport of thyroid and retinol. Beta-glucuronidase increases, depositing estrogen in cells. (A. J.

Cross, et al., "Cortical neurochemistry in Alzheimer-type dementia," Chapter 10, pages 153-170 in Aging of the Brain and Alzheimer's Disease, Prog. in Brain Res. 70, edited by D. F. Swaab, et al., Elsevier, N.Y., 1986.)

3) Abnormally phosphorylated (tau) proteins; association with the variant form of Apo E; tau microtubule organizing proteins, microtubules are involved in transporting cholesterol; phosphorylation, by the kinase systems, regulated by PUFA; the intermediate filaments are generally stress-associated.

4) ApoE, in cytoplasm, involved in cholesterol delivery for pregnenolone synthesis, as in the adrenal; its expression regulated by thyroid. Regulation of the side-chain cleaving enzymes; regulation of the cholesterol intake and conversion to pregnenolone by the endozepine receptor/GABA receptor, modified by progesterone.

AN EXAMPLE OF A REGULATORY PROBLEM

Vegetable oil suppresses the thyroid, increasing estrogen. Estrogen and calcium depolymerize microtubules. Microtubule transport for Apo E, transthyretin, thyroid, and cholesterol for pregnenolone synthesis is disrupted. Transthyretin and Apo E accumulate unused, and deposit in blood vessels, around nerves, and in cytoplasm. Pregnenolone and progesterone deficiency (aggravating thyroid deficiency) causes memory loss, destabilization of nerve cells, failure of myelin formation, and excess cortisol synthesis. Free radicals and calcium cause multiple cell injuries including nerve-death. Estrogen is released by elevated beta-glucuronidase. Imbalances of other steroids, including cortisol and DHEA, develop as cells compensate for pregnenolone deficiency, causing shifts in balance of glial cells. Hypothyroidism, estrogen excess, free unsaturated fats cause increased vascular permeability and brain edema, protein leakage, and alteration of the matrix..

VIII: STRUCTURE AS A REGULATORY SYSTEM--AN EMERGING VISION OF PERVASIVE EPIGENESIS

In the introduction I mentioned that membranous regulation and genetic determination should be considered as defunct theories. What I have been saying about self-actualizing systems and the factors that disrupt them derives from a view of cell function that has been developing since the 1920s.

Around 1940, a Russian biochemist (Oparin, I think) proposed that the enzymes of glycolysis were bound to the structure of the cell when they were not in use, and that they were "desorbed" under the conditions that required abundant glycolysis. Knowing that concept, in 1970 I proposed that the cell water itself underwent a transition under such conditions (which could include increased temperature, reduced oxygen, or nervous or hormonal stimulation). Activation of glycolysis is usually explained by the availability of regulatory substances such as ammonia, phosphate, and NAD, and many biochemists were content to understand cells in terms of test-tube models. But in the last few years, it has become clear that some of these basic regulatory molecules do bind to structural components of the cell. (T. Henics, "Thoughts over cell biology: A commentary," Physiol. Chem. Phys. & Med. NMR 27, 139-140, 1995.) Although the details aren't clear, it is known that hormones and other factors stabilize or destabilize RNA, and that during some of these events relevant enzymes bind to the RNA. When these facts are combined with the information that is accumulating on splicing and modification of RNA, and the copying of RNA back into DNA, the hereditary system is seen to be much more flexible than it was believed to be.

A global change of state is able to steer each part of the process, continuously. In this way, the cell resembles an analog, rather than a digital, control system: each part is momentarily guided, rather than waiting for "feedback."

Where before, cellular "regulatory mechanisms" referred to certain feedback mechanisms based on interactions of randomly diffusing molecules, the new understanding of the cell sees a highly structured system in which very little is random, and the cell's adaptive possibilities, instead of being limited to a certain number of genetic switches, are shaped by every imaginable environmental influence. The cell's structure, far from being "read out of the genome," is sensitively reshaped constantly by processes that incorporate some of the environment in establishing each new stability. The old-model-geneticists have been forced to admit that the genes can't specify everything in the organism's structure, and it was the brain's complexity that forced this recognition that certain things are developed "epigenetically." But the new fact that most biologists are reluctant to accept is that the structure of the cell itself is developed very largely on the basis of information received from the environment--that is, "epigenetically."

Traditionally, epigenesis has meant that the form of an embryo or organism didn't preexist, or wasn't completely specified by the genes. That is, it has had to do with the relationships between cells. It involved a recognition that "cells are clever enough to design an organism." It is a significant step beyond that to the recognition that "cells are clever enough to redesign themselves to meet situations never seen before. "

Biologists working with bacteria and yeasts have seen them adapt in non-random ways to novel conditions. "Directed mutations" are impossible, according to the "central dogma" that has the support of textbooks and most biology professors, but they do occur in those single-celled organisms. Barbara McClintock showed that in corn her mobile genes were mobilized by stress. Although this isn't exactly "directed mutation," it is an example of a mechanism for increasing adaptation when adaptation is need. There is a certain type of enzyme which makes specific cuts in the DNA chain. Biotechnologists find them convenient for their purposes, but their presence serves physiological purposes, presumably in all organisms, like those described by McClintock in corn. During the terminal stress that produces the special kind of cell death known as apoptosis, these enzymes make confetti of the genome.

Poisons, such as estrogen, unsaturated fatty acids, or even radiation, produce different effects at different doses. Low doses typically stimulate cell division, larger doses produce changes of cell type and altered states of differentiation, and finally, adequate doses produce apoptotic cell death. There is a special ideology around apoptosis, which holds that it is "genetically programmed," implying that whenever it occurs in the brain, it was destined to happen sooner or later. But in fact, "growth factors" of various sorts can prevent it. It is increasingly clear that it represents excessive stress and deficient resources. The involvement of the genetic apparatus in differentiation and radical adaptation suggests that the (epigenetic) resources of cells are unlimited.

The changes that are known to be produced by the poisons that we are habitually exposed to are exactly the changes that occur in the aging brain. As I scan over hundreds of studies that define the effects of estrogen, unsaturated fats, excess iron, and lipid peroxidation, my argument seems commonplace, even trivial, except that I know that it clearly relates to therapies for most of the degenerative diseases, and that the great culture-machine is propagating a different view at several points that are essential for my argument.

They are advancing a myth about human nature, so I will advance a counter-myth. At the time people were growing their large brains they lived in the tropics. I suggest that in this time before the development of grain-based agriculture, they ate a diet that was relatively free of unsaturated fats and low in iron--based on tropical fruits. I suggest that the Boskop skull from Mt. Kilimanjaro was representative of people under those conditions, and that just by our present knowledge of the association of brain size with longevity, they--as various "Golden Age" myths claim--must have had a very long life-span. As people moved north and developed new ways of living, their consumption of unsaturated fats increased, their brain size decreased, and they aged rapidly. Neanderthal relics show that flaxseed was a staple of their diet.

Even living in the tropics, there are many possibilities for diets rich in signal-disrupting substances, including iron, and in high latitudes there are opportunities for reducing our exposure to them. As a source of protein, milk is uniquely low in its iron content. Potatoes, because of the high quality of their protein, are probably relatively free of toxic signal-substances. Many tropical fruits, besides having relatively saturated fats, are also low in iron, and often contain important quantities of amino acids and proteins. In this context, Jeanne Calment's life-long, daily consumption of chocolate comes to mind: As she approaches her 121st birthday, she is still eating chocolate, though she has stopped smoking and drinking wine. The saturated fats in chocolate have been found to block the toxicity of oils rich in linoleic acid, and its odd proteins seem to have an anabolic action.

If we really take seriously even the traditional sort of epigenesis, and especially if we accept the deeper idea of epigenesis on the level of cellular structure and function, we have to see the organism as a sort of "whirlwind of cells," made up of whirlwinds of atoms (in Vernadsky's phrase) in which our way of life sets the boundaries within which our cells will restructure themselves.

The random production of free radicals, rather than acting only by way of genetic damage or protein cross-linking, is also able to act as a signalling process, that is, on a strictly physiological level. An excess of unsaturated fatty acids itself constitutes a massive distortion of the regulatory systems, but it also leads to distortions in the "eicosanoid" system and the increasingly uncontrolled production of free radicals, and to changes in energy, thyroid activity, and steroid balance. The aging body, rather than being like a car that needs more and more repairs until it collapses from simple wear, is more like a car traveling a road that becomes increasingly rough and muddy, until the road becomes an impassable swamp.

The suggested therapy is a correction of the signalling process, rather than "genetic surgery," transplantation, etc., which is the pessimistic implication of the doctrine that oxidative damage is simply a matter of "wear and tear," "somatic mutations," and "cross-linking." Those problems are reparable, and our emphasis should be on the production of energy and the avoidance of the conditions that allow the undesirable signals to accumulate.

The absence of cancer on a diet lacking unsaturated fats, the increased rate of metabolism, decreased free radical production, resistance to stress and poisoning by iron, alcohol, endotoxin, alloxan and streptozotocin, etc., improvement of brain structure and function, decreased susceptibility to blood clots, and lack of obesity and age pigment on a diet using coconut oil rather than unsaturated fats, indicates that something very simple can be done to reduce the suffering from the major degenerative diseases, and that it is very likely acting by reducing the aging process itself at its physiological core.

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